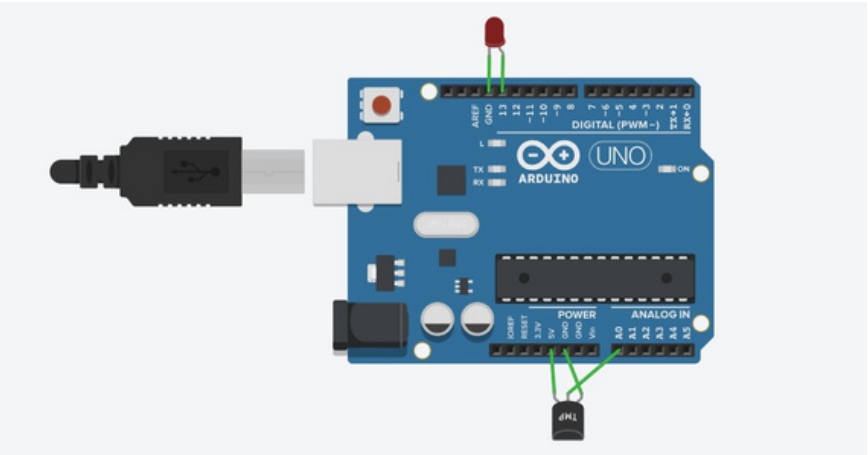
2. Write a program for sending alert messages to the user for controlling and interacting with your environment.

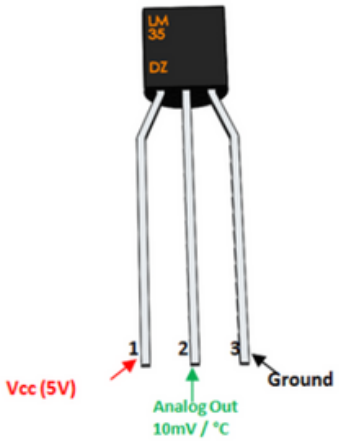
Code :

```
const int sensorPin = A0;
const int ledPin = 13;
const float tempThreshold = 24.50;  // Temperature in Celsius to trigger alert

void setup() {
  pinMode(ledPin, OUTPUT);
  Serial.begin(9600);
}
float readTemperatureC() {
  int sensorValue = analogRead(sensorPin);
  float voltage = sensorValue * (5.0 / 1023.0); // Convert ADC value to voltage
  float temperatureC = (voltage - 0.5) * 100.0; // TMP36 conversion formula
  return temperatureC;
}
void loop() {
  float temperature = readTemperatureC();
  Serial.print("Temperature: ");
  Serial.print(temperature);
  Serial.println(" °C");

  if (temperature > tempThreshold) {
    digitalWrite(ledPin, HIGH);  // Turn LED on
    Serial.println("ALERT! Temperature too high!");
  } else {
    digitalWrite(ledPin, LOW);   // Turn LED off
  }
  delay(1000);
}
```

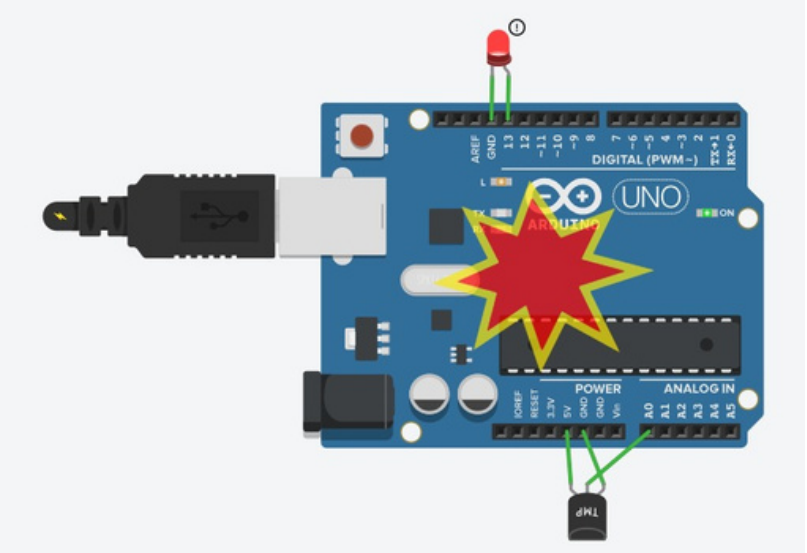
Circuit Diagram :

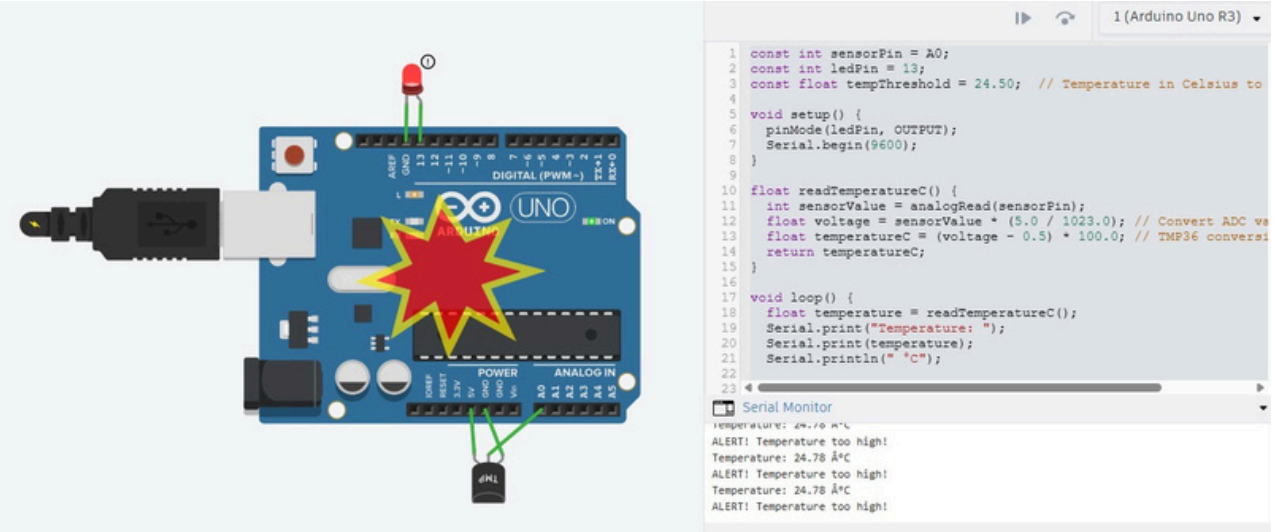


Temp Sensor Pin Diagram :



Output :


DIGITAL (PWM~)
UNO
POWER
ANALOG IN
TMP


1 (Arduino Uno R3)
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float temperatureC = (voltage - 0.5) * 100.0; // TMP36 conversi
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}
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Serial.print(temperature);
Serial.println(" °C");
Serial Monitor
ALERT! Temperature too high!
Temperature: 24.78 Â°C
ALERT! Temperature too high!
Temperature: 24.78 Â°C
ALERT! Temperature too high!